

SECTION B

6 (short)

- (a) Prove de Morgan's theorems for two variables:

$$\overline{A.B} = \overline{A} + \overline{B} \quad \text{and} \quad \overline{A+B} = \overline{A}.\overline{B} \quad [4]$$

- (b) Consider the Boolean expression for Z:

$$Z = B.C + A.\overline{B} + \overline{A}.\overline{B}$$

By constructing a Karnaugh map for $\overline{Z}$, or otherwise, find a simplified expression for Z that can be implemented by two-input NAND gates only, and draw a circuit diagram that implements the expression. [6]

7 (short) A certain memory chip has 13 address lines and 8 data lines. It is connected to a microprocessor with a 16 bit address bus and an 8 bit data bus.

- (a) Determine the capacity of the memory chip in bytes, and compare its capacity to the maximum amount of memory that a modern microprocessor with a 32 bit address bus and an 8 bit data bus can address. [3]

- (b) Explain the roles of the $\overline{CS}$ and $R/\overline{W}$ lines in accessing data stored in the memory chip, and determine a Boolean expression for the $\overline{CS}$ input such that the memory chip appears in the microprocessor's memory map between locations A000_H and BFFF_H. Hence draw a circuit showing how the memory chip should be connected to the microprocessor. [7]